

ApexPlanet Software Pvt. Ltd.

INTERNSHIP TASK REPORT

Network Security & Scanning

Domain: Cybersecurity & Ethical Hacking

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1. Objective

The objective of this task was to gain hands-on, practical exposure to core network security and vulnerability scanning techniques used in real-world cybersecurity assessments. This involved setting up and operating industry-standard tools to discover live hosts and open services, capture and analyse network traffic, identify vulnerabilities using an automated scanner, simulate network stress/flood conditions in a controlled manner, and enumerate web application directories to understand the exposed attack surface of a target system. The exercise was aimed at strengthening the practical understanding of reconnaissance, scanning, and monitoring phases that form the foundation of ethical hacking and penetration testing workflows.

2. Tools Used

Tool	Purpose in this Task
Nmap	Network scanning tool used for host discovery, open port identification, service/version detection, and OS fingerprinting.
Wireshark	Packet capture and protocol analysis tool used to record and inspect live network traffic on wireless and Ethernet interfaces.
OpenVAS / GVM	Greenbone Vulnerability Management suite used to perform automated vulnerability assessment against the target host.
hping3	Command-line packet crafting tool used to simulate SYN-based traffic and test port responsiveness under a controlled flood scenario.
Gobuster	Directory/file brute-forcing tool used to enumerate hidden or unlisted paths and endpoints on a target web server.
iptables	Linux firewall utility referenced for configuring host-based packet filtering rules to control and restrict inbound/outbound traffic.

3. Methodology

3.1 Network & Host Reconnaissance

- Performed a TCP SYN scan (nmap -sS) on the target host to identify open ports without completing the full TCP handshake.
- Executed OS fingerprinting (nmap -O) to determine the operating system and device type of the target.
- Ran a service/version detection scan and a UDP scan (-sU) to gather deeper service-level information.
- Used nc -v and nslookup to independently verify port reachability and DNS resolution.

3.2 Vulnerability Assessment with OpenVAS/GVM

- Installed the Greenbone Vulnerability Management (GVM) stack including gvm, gvm-tools, gvm-tools, and gvm-tools.

- Created and configured an administrative GVM user account and reset credentials where required.
- Synchronised vulnerability feed data (NVT, SCAP, CERT) and resolved feed-lock and service-startup issues.
- Validated the installation using gvm-check-setup and started the GVM services (gvmd, gsad, ospd-openvas).

3.3 Traffic Capture and Analysis with Wireshark

- Captured live traffic on the wlan0, eth0, and loopback interfaces during scanning activity.
- Saved capture sessions as .pcapng files for later inspection of packet-level behaviour.

3.4 Controlled Flood Simulation with hping3

- Verified hping3 installation and crafted raw SYN packets toward the target host.
- Executed a flood-mode SYN test on port 80 to observe traffic generation at scale in a controlled lab environment.
- Sent targeted SYN probes to ports 22 and 80 individually to record RTT and response-flag statistics.

3.5 Web Directory Enumeration with Gobuster

- Reviewed the Gobuster help menu to understand available enumeration modes (dir, vhost, dns, fuzz).
- Corrected initial command-syntax and wordlist-path errors encountered during setup.
- Executed a directory enumeration scan using the dirb common.txt wordlist against the target domain.
- Filtered results for security-sensitive keywords (admin, config, password, api, backup) and verified select endpoints with curl.

4. Findings / Results

The following screenshots document the practical execution of each phase described above, captured directly from the terminal during the task.

```

ospd-openvas.service - OSPd Wrapper for the OpenVAS Scanner (ospd-openvas)
Loaded: loaded (/usr/lib/systemd/system/ospd-openvas.service; disabled; preset: disabled)
Active: active (running) since Wed 2026-09-23 14:45:06 IST; 5s ago
Invocation: 2307b6427fa048edaa0abe8c6f5bb9f4
Docs: man:ospd-openvas(8)
man:openvas(8)
Process: 58743 ExecStart=/usr/bin/ospd-openvas --config /etc/gvm/ospd-openvas.conf --log-config /etc/gvm/ospd-logging.conf (code=exited, status=0/SUCCESS)
Main PID: 58753 (ospd-openvas)
Tasks: 7 (limit: 18072)
Memory: 128.7M (peak: 136.8M)
CPU: 1.062s
CGroup: /system.slice/ospd-openvas.service
└─58753 /usr/bin/python3 /usr/bin/ospd-openvas --config /etc/gvm/ospd-openvas.conf --log-config /etc/gvm/ospd-logging.conf
└─58755 /usr/bin/python3 -c "from multiprocessing.resource_tracker import main;main(9)"
└─58756 /usr/bin/python3 -c "import sys; from multiprocessing.forkserver import main; main(9, 11, ['__main__', sys_argv=sys.argv[1:]], **{'sys_path': ['/usr/bin', '/usr/lib/python3.14/zlib-dynload', '/usr/local/lib/python3.14/dist-packages', '/usr/lib/python3.14/dist-packages', '/usr/lib/python3.14/dist-packages', 'main_path': '/usr/bin/ospd-openvas', 'authkey_r': 13])) /usr
-openvas.conf --log-config /etc/gvm/ospd-logging.conf
└─58762 /usr/bin/python3 -c "import sys; from multiprocessing.forkserver import main; main(9, 11, ['__main__', sys_argv=sys.argv[1:]], **{'sys_path': ['/usr/bin', '/usr/lib/python3.14/zlib-dynload', '/usr/local/lib/python3.14/dist-packages', '/usr/lib/python3.14/dist-packages', '/usr/lib/python3.14/dist-packages', 'main_path': '/usr/bin/ospd-openvas', 'authkey_r': 13])) /usr
-openvas.conf --log-config /etc/gvm/ospd-logging.conf

Sep 23 14:45:05 kali systemd[1]: Starting ospd-openvas.service - OSPd Wrapper for the OpenVAS Scanner (ospd-openvas)...
Sep 23 14:45:06 kali systemd[1]: Started ospd-openvas.service - OSPd Wrapper for the OpenVAS Scanner (ospd-openvas).

[>] Opening Web UI (https://127.0.0.1:9392) in: 5... 4... 3... 2... 1...

(root@kali) ~/home/akshat
# gvm --get-users

(root@kali) ~/home/akshat
# systemctl status gvm

gvm.service - Greenbone Vulnerability Manager daemon (gvm)
Loaded: loaded (/usr/lib/systemd/system/gvm.service; disabled; preset: disabled)
Active: active (running) since Wed 2026-09-23 14:45:07 IST; 2min 5s ago
Invocation: 130139134daa42efb72374b967e24585
Docs: man:gvm(8)
Process: 58757 ExecStart=/usr/sbin/gvm --osp-vt-update=/run/ospd/ospd-openvas.sock --listen-groups=gvm (code=exited, status=0/SUCCESS)
Main PID: 58759 (gvm)
Tasks: 3 (limit: 18072)
Memory: 6.2M (peak: 10.7M)
CPU: 235ms
CGroup: /system.slice/gvm.service
└─58759 "gvm: Waiting " --osp-vt-update=/run/ospd/ospd-openvas.sock --listen-groups=gvm
└─58789 "gvm: Synchron " --osp-vt-update=/run/ospd/ospd-openvas.sock --listen-groups=gvm
└─58796 "gvm: Syncing " --osp-vt-update=/run/ospd/ospd-openvas.sock --listen-groups=gvm

Sep 23 14:45:06 kali systemd[1]: Starting gvm.service - Greenbone Vulnerability Manager daemon (gvm)...
Sep 23 14:45:06 kali systemd[1]: gvm.service: Can't open PID file '/run/gvm/gvm.pid' (yet?) after start: No such file or directory
Sep 23 14:45:07 kali systemd[1]: Started gvm.service - Greenbone Vulnerability Manager daemon (gvm).

(root@kali) ~/home/akshat
# gvm --get-users --verbose

```

Figure 1: OSPd-OpenVAS and GVM Service Status

Output of `systemctl status` for `ospd-openvas.service` and `gvm.service`, confirming that the OpenVAS scanner wrapper and the Greenbone Vulnerability Manager daemon were successfully started and are in the active (running) state, along with the `gvm --get-users` command used to verify configured GVM users.

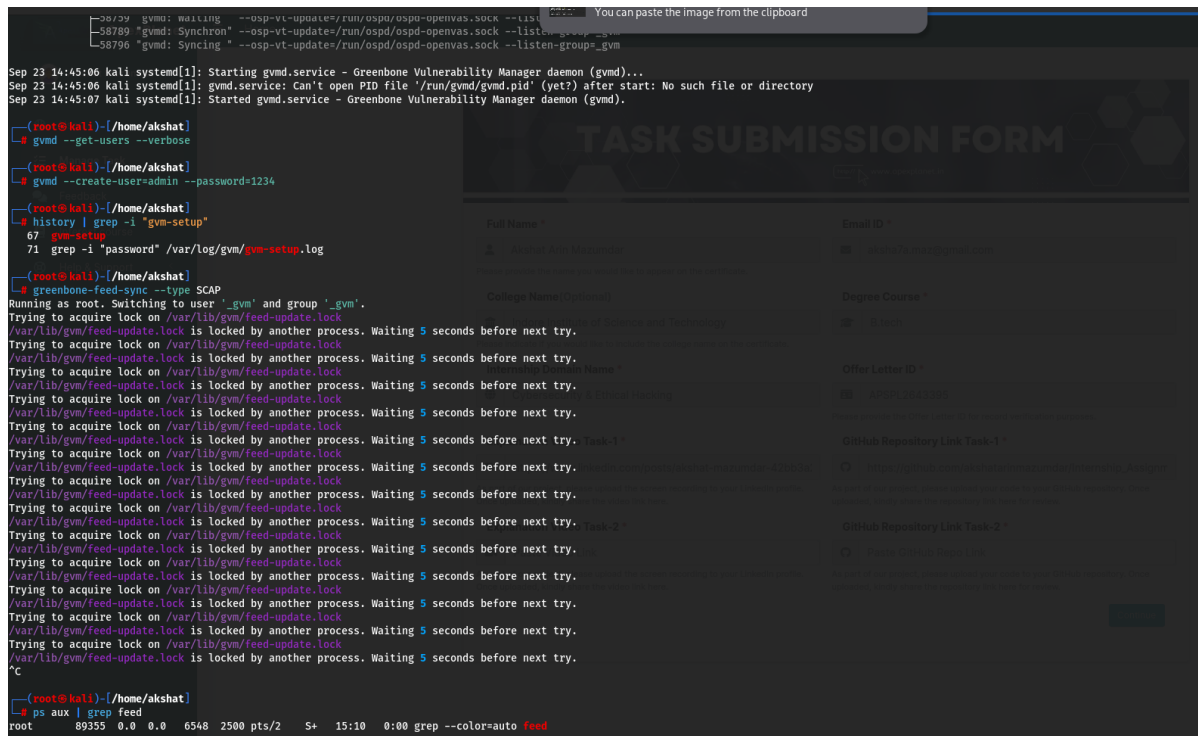


Figure 2: GVM User Creation and Feed Synchronisation Attempt

Creation of an administrative GVM user (`gvmc --create-user=admin`) and an initial attempt to run `greenbone-feed-sync` to update SCAP vulnerability data. The `feed-update.lock` file was held by another background process, so the sync repeatedly waited before retrying.

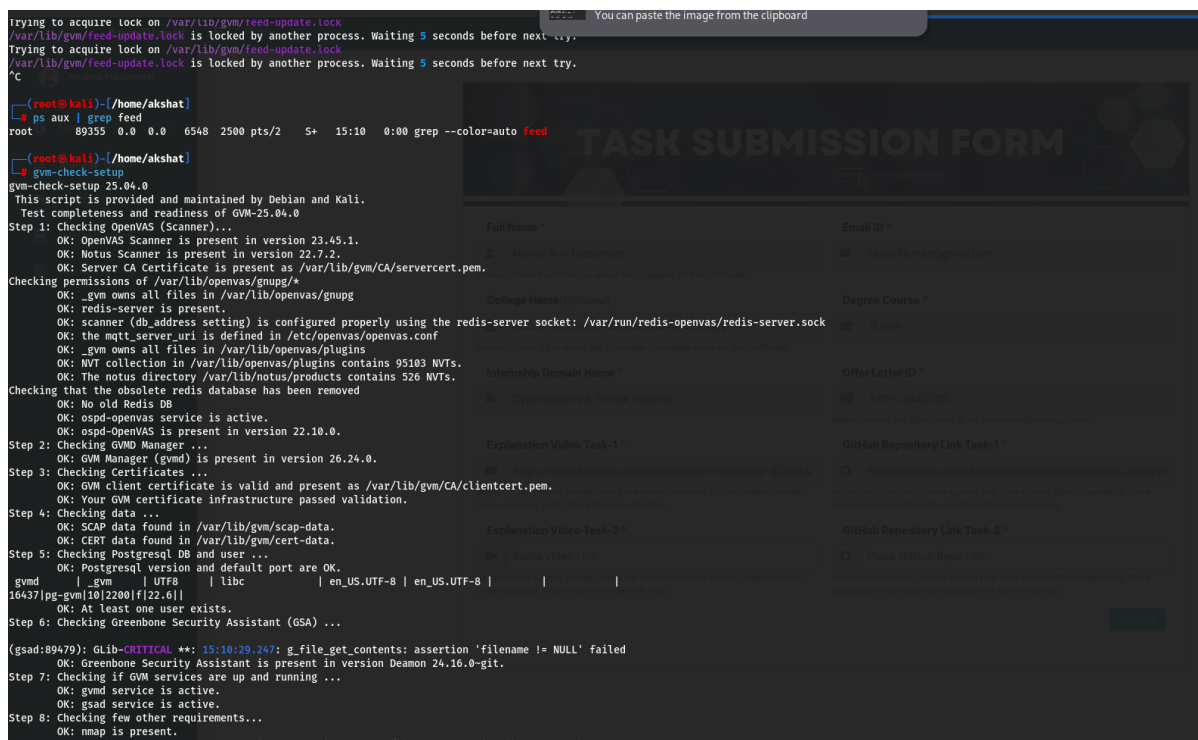


Figure 3: Feed Lock Retry and GVM Setup Verification (Part 1)

Continuation of the feed-update lock retry loop, followed by the gym-check-setup diagnostic utility, which validates the OpenVAS scanner, NVT/Notus data, certificates, and database configuration to confirm the installation is healthy.

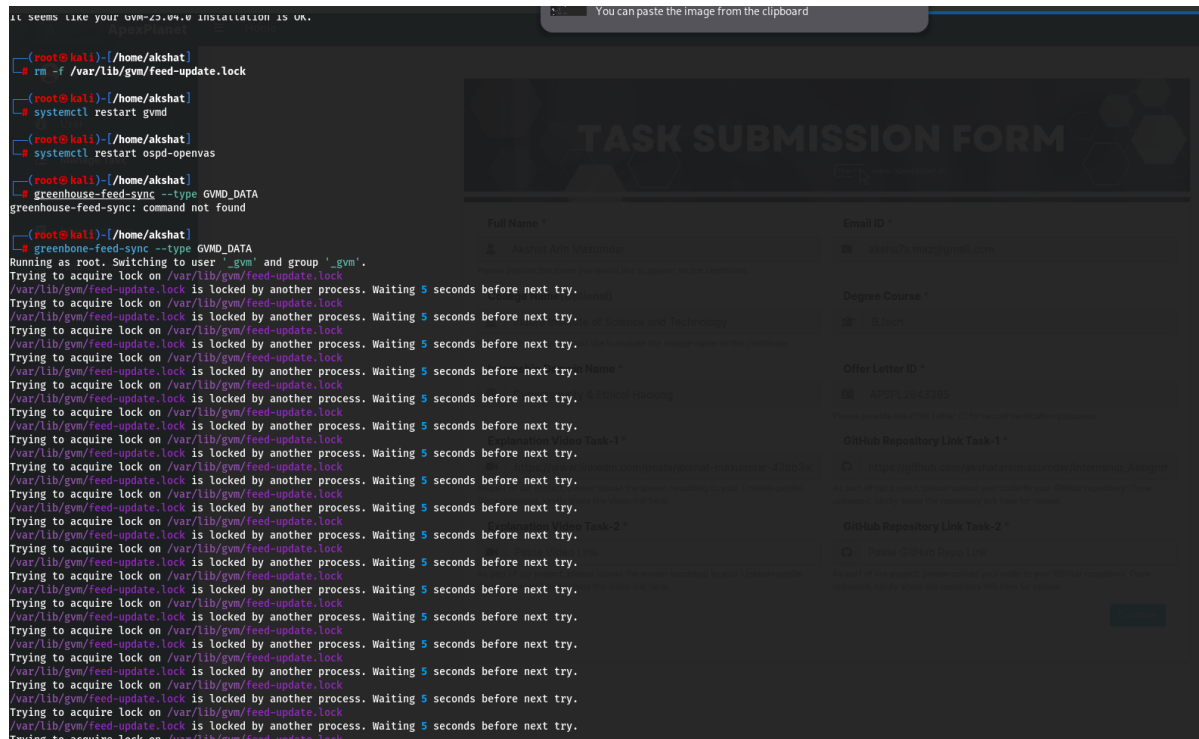


Figure 4: Manual Lock Removal and Service Restart

The stale feed-update.lock file was manually removed and the gvmd and ospd-openvas services were restarted to clear the blocked state before re-attempting the GVMD data feed synchronisation.

```

OK: gvm service is active.
OK: gsd service is active.
Step 8: Checking few other requirements...
OK: nmap is present.
OK: ssh-keygen found, LSC credential generation for GNU/Linux targets is likely to work.
OK: nsis found, LSC credential package generation for Microsoft Windows targets is likely to work.
OK: SELinux is disabled.
OK: xsltproc found.
WARNING: Your password policy is empty.
SUGGEST: Edit the /etc/gvm/pwpolicy.conf file to set a password policy.
Step 9: Checking greenbone-security-assistant...
OK: greenbone-security-assistant is installed

It seems like your GVM-25.04.0 installation is OK.

(root@kali) ~/home/akshat
# gvm-start
[1] GVM services are already running

(root@kali) ~/home/akshat
# systemctl restart gvm

(root@kali) ~/home/akshat
# u_gvm gvm --get-configs
-u: command not found

(root@kali) ~/home/akshat
# wireshark
** (Wireshark:99621) 15:26:26.016476 [Capture MESSAGE] -- Capture Start ...
** (Wireshark:99621) 15:26:26.063317 [Capture MESSAGE] -- Capture started
** (Wireshark:99621) 15:26:26.063368 [Capture MESSAGE] -- File: "/tmp/wireshark_wlan0K5ZV3.pcapng"
** (Wireshark:99621) 15:26:26.363190 [GUI WARNING] -- dbus reply error: [ "org.freedesktop.DBus.Error.NoReply" ] "Did not receive a reply. Possible causes include: the remote application did not se
icy blocked the reply, the reply timeout expired, or the network connection was broken."
** (Wireshark:99621) 15:26:26.363267 [GUI WARNING] -- dbus reply error: [ "org.freedesktop.DBus.Error.NoReply" ] "Did not receive a reply. Possible causes include: the remote application did not se
icy blocked the reply, the reply timeout expired, or the network connection was broken."
** (Wireshark:99621) 15:26:26.363318 [GUI WARNING] -- Failed to register with host portal QDBusError("org.freedesktop.DBus.Error.NoReply", "Did not receive a reply. Possible causes include: the rem
e message bus security policy blocked the reply, the reply timeout expired, or the network connection was broken.")
** (Wireshark:99621) 15:27:09.608395 [Capture MESSAGE] -- Capture Stop ...
** (Wireshark:99621) 15:27:09.658865 [Capture MESSAGE] -- Capture stopped.
** (Wireshark:99621) 15:28:10.216004 [Capture MESSAGE] -- Capture Start ...
** (Wireshark:99621) 15:28:10.442398 [Capture MESSAGE] -- Capture started
** (Wireshark:99621) 15:28:10.442426 [Capture MESSAGE] -- File: "/tmp/wireshark_eth0P8M5V3.pcapng"
** (Wireshark:99621) 15:29:34.951998 [Capture MESSAGE] -- Capture Stop ...
** (Wireshark:99621) 15:29:34.998600 [Capture MESSAGE] -- Capture stopped.
** (Wireshark:99621) 15:29:35.003466 [Capture MESSAGE] -- Capture Start ...
** (Wireshark:99621) 15:29:35.234905 [Capture MESSAGE] -- Capture started
** (Wireshark:99621) 15:29:35.234946 [Capture MESSAGE] -- File: "/tmp/wireshark_eth0KAQYV3.pcapng"
** (Wireshark:99621) 15:29:38.250841 [Capture MESSAGE] -- Capture Stop ...
** (Wireshark:99621) 15:29:38.310750 [Capture MESSAGE] -- Capture stopped.
** (Wireshark:99621) 15:30:54.171308 [Capture MESSAGE] -- Capture Start ...
** (Wireshark:99621) 15:30:54.202778 [Capture MESSAGE] -- Capture started
** (Wireshark:99621) 15:30:54.202835 [Capture MESSAGE] -- File: "/tmp/wireshark_LoopbackJXGZV3.pcapng"
** (Wireshark:99621) 15:32:05.266828 [Capture MESSAGE] -- Capture Stop ...

```

Figure 5: GVM Setup Completion and Wireshark Capture Start

Final steps of gvm-check-setup confirming a healthy GVM-25.04.0 installation ('installation is OK'), followed by starting the GVM web services (gvm-start) and launching Wireshark to begin capturing live traffic on the wlan0 and eth0 interfaces.

```

** (Wireshark:99621) 15:30:54.171308 [Capture MESSAGE] -- Capture Start ...
** (Wireshark:99621) 15:30:54.202778 [Capture MESSAGE] -- Capture started
** (Wireshark:99621) 15:30:54.202835 [Capture MESSAGE] -- File: "/tmp/wireshark_LoopbackJXGZV3.pcapng"
** (Wireshark:99621) 15:33:05.356858 [Capture MESSAGE] -- Capture Stop ...
** (Wireshark:99621) 15:33:05.415089 [Capture MESSAGE] -- Capture stopped.
** (Wireshark:99621) 15:33:05.428883 [Capture MESSAGE] -- Capture Start ...
** (Wireshark:99621) 15:33:05.495749 [Capture MESSAGE] -- Capture started
** (Wireshark:99621) 15:33:05.495827 [Capture MESSAGE] -- File: "/tmp/wireshark_LoopbackZ5A4V3.pcapng"
** (Wireshark:99621) 15:33:33.992006 [Capture MESSAGE] -- Capture Stop ...
** (Wireshark:99621) 15:33:34.050901 [Capture MESSAGE] -- Capture stopped.
** (Wireshark:99621) 15:33:34.146154 [Capture MESSAGE] -- Capture Start ...
** (Wireshark:99621) 15:33:34.239944 [Capture MESSAGE] -- Capture started
** (Wireshark:99621) 15:33:34.239996 [Capture MESSAGE] -- File: "/tmp/wireshark_Loopback54CZV3.pcapng"
** (Wireshark:99621) 15:33:35.416660 [Capture MESSAGE] -- Capture Stop ...
** (Wireshark:99621) 15:33:35.475135 [Capture MESSAGE] -- Capture stopped.

(root@kali) ~/home/akshat
# apt install hping3 -y
hping3 is already the newest version (3.a2.ds2-11-kali1).
hping3 set to manually installed.
The following packages were automatically installed and are no longer required:
  libgsoap-2.8.139 libpoppler147 libpython3.13-dev python3.13-dev python3.13-minimal python3.13-venv
Use 'sudo apt autoremove' to remove them.

Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 473

(root@kali) ~/home/akshat
# hping3 -S --flood -p 80 10.16.177.22

HPING 10.16.177.22 (wlan0 10.16.177.22): S set, 40 headers + 0 data bytes
hping in flood mode, no replies will be shown
^[[^C
--- 10.16.177.22 hping statistic ---
5535055 packets transmitted, 0 packets received, 100% packet loss
round-trip min/avg/max = 0.0/0.0/0.0 ms

(root@kali) ~/home/akshat
# hping3 -S -p 22 10.16.177.22

HPING 10.16.177.22 (wlan0 10.16.177.22): S set, 40 headers + 0 data bytes
len=40 ip=10.16.177.22 ttl=64 DF id=0 sport=22 flags=RA seq=0 win=0 rtt=3.3 ms
len=40 ip=10.16.177.22 ttl=64 DF id=0 sport=22 flags=RA seq=1 win=0 rtt=7.2 ms
len=40 ip=10.16.177.22 ttl=64 DF id=0 sport=22 flags=RA seq=2 win=0 rtt=15.1 ms
len=40 ip=10.16.177.22 ttl=64 DF id=0 sport=22 flags=RA seq=3 win=0 rtt=2.9 ms
len=40 ip=10.16.177.22 ttl=64 DF id=0 sport=22 flags=RA seq=4 win=0 rtt=20.0 ms
len=40 ip=10.16.177.22 ttl=64 DF id=0 sport=22 flags=RA seq=5 win=0 rtt=7.7 ms
len=40 ip=10.16.177.22 ttl=64 DF id=0 sport=22 flags=RA seq=6 win=0 rtt=7.6 ms
len=40 ip=10.16.177.22 ttl=64 DF id=0 sport=22 flags=RA seq=7 win=0 rtt=6.5 ms
len=40 ip=10.16.177.22 ttl=64 DF id=0 sport=22 flags=RA seq=8 win=0 rtt=4.3 ms
len=40 ip=10.16.177.22 ttl=64 DF id=0 sport=22 flags=RA seq=9 win=0 rtt=2.2 ms
len=40 ip=10.16.177.22 ttl=64 DF id=0 sport=22 flags=RA seq=10 win=0 rtt=6.1 ms
len=40 ip=10.16.177.22 ttl=64 DF id=0 sport=22 flags=RA seq=11 win=0 rtt=3.0 ms

```

Figure 6: Wireshark Loopback Capture and hping3 SYN Flood Test

Wireshark capture logs for the loopback interface, followed by verification that hping3 was installed and a controlled SYN flood simulation launched against the target host (10.16.177.22) on port 80 in flood mode to observe network behaviour under high packet-rate conditions.

```

--- 10.16.177.22 hping statistic ---
43 packets transmitted, 43 packets received, 0% packet loss
round-trip min/avg/max = 1.6/13.0/106.2 ms

--(root@kali)-[/home/akshat]
# hping3 -S -p 80 10.16.177.22

HPING 10.16.177.22 (wlan0 10.16.177.22): 5 set, 40 headers + 0 data bytes
len=40 ip=10.16.177.22 ttl=64 DF id=0 sport=80 flags=RA seq=0 win=0 rtt=64.2 ms
len=40 ip=10.16.177.22 ttl=64 DF id=0 sport=80 flags=RA seq=1 win=0 rtt=8.1 ms
len=40 ip=10.16.177.22 ttl=64 DF id=0 sport=80 flags=RA seq=2 win=0 rtt=17.0 ms
len=40 ip=10.16.177.22 ttl=64 DF id=0 sport=80 flags=RA seq=3 win=0 rtt=81.8 ms
len=40 ip=10.16.177.22 ttl=64 DF id=0 sport=80 flags=RA seq=4 win=0 rtt=6.7 ms
len=40 ip=10.16.177.22 ttl=64 DF id=0 sport=80 flags=RA seq=5 win=0 rtt=33.5 ms
len=40 ip=10.16.177.22 ttl=64 DF id=0 sport=80 flags=RA seq=6 win=0 rtt=69.4 ms
len=40 ip=10.16.177.22 ttl=64 DF id=0 sport=80 flags=RA seq=7 win=0 rtt=16.3 ms
len=40 ip=10.16.177.22 ttl=64 DF id=0 sport=80 flags=RA seq=8 win=0 rtt=9.2 ms
len=40 ip=10.16.177.22 ttl=64 DF id=0 sport=80 flags=RA seq=9 win=0 rtt=6.0 ms
len=40 ip=10.16.177.22 ttl=64 DF id=0 sport=80 flags=RA seq=10 win=0 rtt=6.9 ms
len=40 ip=10.16.177.22 ttl=64 DF id=0 sport=80 flags=RA seq=11 win=0 rtt=6.8 ms
len=40 ip=10.16.177.22 ttl=64 DF id=0 sport=80 flags=RA seq=12 win=0 rtt=5.7 ms
len=40 ip=10.16.177.22 ttl=64 DF id=0 sport=80 flags=RA seq=13 win=0 rtt=5.6 ms
len=40 ip=10.16.177.22 ttl=64 DF id=0 sport=80 flags=RA seq=14 win=0 rtt=6.4 ms
len=40 ip=10.16.177.22 ttl=64 DF id=0 sport=80 flags=RA seq=15 win=0 rtt=6.2 ms
len=40 ip=10.16.177.22 ttl=64 DF id=0 sport=80 flags=RA seq=16 win=0 rtt=44.2 ms
len=40 ip=10.16.177.22 ttl=64 DF id=0 sport=80 flags=RA seq=17 win=0 rtt=6.1 ms
len=40 ip=10.16.177.22 ttl=64 DF id=0 sport=80 flags=RA seq=18 win=0 rtt=7.0 ms
len=40 ip=10.16.177.22 ttl=64 DF id=0 sport=80 flags=RA seq=19 win=0 rtt=30.8 ms
len=40 ip=10.16.177.22 ttl=64 DF id=0 sport=80 flags=RA seq=20 win=0 rtt=5.7 ms
len=40 ip=10.16.177.22 ttl=64 DF id=0 sport=80 flags=RA seq=21 win=0 rtt=5.6 ms
len=40 ip=10.16.177.22 ttl=64 DF id=0 sport=80 flags=RA seq=22 win=0 rtt=44.5 ms
len=40 ip=10.16.177.22 ttl=64 DF id=0 sport=80 flags=RA seq=23 win=0 rtt=5.4 ms
len=40 ip=10.16.177.22 ttl=64 DF id=0 sport=80 flags=RA seq=24 win=0 rtt=5.3 ms
len=40 ip=10.16.177.22 ttl=64 DF id=0 sport=80 flags=RA seq=25 win=0 rtt=8.2 ms
len=40 ip=10.16.177.22 ttl=64 DF id=0 sport=80 flags=RA seq=26 win=0 rtt=7.1 ms
len=40 ip=10.16.177.22 ttl=64 DF id=0 sport=80 flags=RA seq=27 win=0 rtt=5.9 ms
len=40 ip=10.16.177.22 ttl=64 DF id=0 sport=80 flags=RA seq=28 win=0 rtt=7.8 ms
len=40 ip=10.16.177.22 ttl=64 DF id=0 sport=80 flags=RA seq=29 win=0 rtt=5.7 ms
len=40 ip=10.16.177.22 ttl=64 DF id=0 sport=80 flags=RA seq=30 win=0 rtt=5.6 ms
len=40 ip=10.16.177.22 ttl=64 DF id=0 sport=80 flags=RA seq=31 win=0 rtt=5.5 ms
len=40 ip=10.16.177.22 ttl=64 DF id=0 sport=80 flags=RA seq=32 win=0 rtt=4.5 ms
len=40 ip=10.16.177.22 ttl=64 DF id=0 sport=80 flags=RA seq=33 win=0 rtt=15.3 ms
len=40 ip=10.16.177.22 ttl=64 DF id=0 sport=80 flags=RA seq=34 win=0 rtt=12.2 ms
len=40 ip=10.16.177.22 ttl=64 DF id=0 sport=80 flags=RA seq=35 win=0 rtt=5.1 ms
len=40 ip=10.16.177.22 ttl=64 DF id=0 sport=80 flags=RA seq=36 win=0 rtt=6.0 ms
len=40 ip=10.16.177.22 ttl=64 DF id=0 sport=80 flags=RA seq=37 win=0 rtt=33.0 ms
len=40 ip=10.16.177.22 ttl=64 DF id=0 sport=80 flags=RA seq=38 win=0 rtt=4.9 ms
len=40 ip=10.16.177.22 ttl=64 DF id=0 sport=80 flags=RA seq=39 win=0 rtt=19.8 ms
^C
--- 10.16.177.22 hping statistic ---
40 packets transmitted, 40 packets received, 0% packet loss
round-trip min/avg/max = 4.5/16.3/81.8 ms

```

Figure 7: hping3 Targeted Port Probing (SSH and HTTP)

Controlled hping3 SYN probes sent to port 22 (SSH) and port 80 (HTTP) on the target host, capturing response flags (RA - Reset/Ack) and round-trip time statistics, which help assess port responsiveness and basic network latency.

```

=====
--(akshat@kali)-[~]
$ gobuster dir -u google.com -w /usr/share/wordlists/dirb/common.txt -o result.txt

=====
Gobuster v3.8.2
by OJ Reeves (@TheColonial) & Christian Mehlmauer (@firefart)
=====
[+] Url:             http://google.com
[+] Method:          GET
[+] Threads:         10
[+] Wordlist:         /usr/share/wordlists/dirb/common.txt
[+] Negative Status codes: 404
[+] User Agent:      gobuster/3.8.2
[+] Timeout:         10s
=====
Starting gobuster in directory enumeration mode
=====
2003 (Status: 301) [Size: 224] [--> https://www.google.com/2003]
2001 (Status: 301) [Size: 224] [--> https://www.google.com/2001]
2002 (Status: 301) [Size: 224] [--> https://www.google.com/2002]
2004 (Status: 301) [Size: 224] [--> https://www.google.com/2004]
2006 (Status: 301) [Size: 224] [--> https://www.google.com/2006]
2005 (Status: 301) [Size: 224] [--> https://www.google.com/2005]
2007 (Status: 301) [Size: 224] [--> https://www.google.com/2007]
2008 (Status: 301) [Size: 224] [--> https://www.google.com/2008]
2010 (Status: 301) [Size: 224] [--> https://www.google.com/2010]
2009 (Status: 301) [Size: 224] [--> https://www.google.com/2009]
2012 (Status: 301) [Size: 224] [--> https://www.google.com/2012]
2014 (Status: 301) [Size: 224] [--> https://www.google.com/2014]
2013 (Status: 301) [Size: 224] [--> https://www.google.com/2013]
2011 (Status: 301) [Size: 224] [--> https://www.google.com/2011]
A (Status: 301) [Size: 221] [--> https://www.google.com/A]
a (Status: 301) [Size: 221] [--> https://www.google.com/a]
about (Status: 301) [Size: 225] [--> https://www.google.com/about]
accessibility (Status: 301) [Size: 233] [--> https://www.google.com/accessibility]
account (Status: 301) [Size: 227] [--> https://www.google.com/account]
accounts (Status: 302) [Size: 237] [--> https://accounts.google.com/]
action (Status: 301) [Size: 225] [--> https://www.google.com/action]
activity (Status: 301) [Size: 0] [--> https://google.com/activity]
ads (Status: 301) [Size: 223] [--> https://www.google.com/ads]
advanced_search (Status: 301) [Size: 234] [--> http://www.google.com/advanced_search]
advertise (Status: 301) [Size: 229] [--> https://www.google.com/advertise]
advertisers (Status: 301) [Size: 231] [--> https://www.google.com/advertisers]
advertising (Status: 301) [Size: 231] [--> https://www.google.com/advertising]
adview (Status: 301) [Size: 225] [--> https://www.google.com/adview]
africa (Status: 301) [Size: 226] [--> https://www.google.com/africa]
alerts (Status: 301) [Size: 226] [--> https://www.google.com/alerts]
analytics (Status: 301) [Size: 229] [--> https://www.google.com/analytics]
android (Status: 301) [Size: 227] [--> https://www.google.com/android]
answers (Status: 301) [Size: 227] [--> https://www.google.com/answers]

```

Figure 8: Gobuster Directory Enumeration - Scan Initialisation

Gobuster launched in directory enumeration mode against google.com using the dirb common.txt wordlist, showing the scan configuration (threads, negative status codes, user agent) and the first set of discovered paths with their HTTP status codes and redirect targets.

```

=====
cr (Status: 400) [Size: 16] [--> https://www.google.com/cr]
dashboard (Status: 301) [Size: 229] [--> https://www.google.com/dashboard]
deleted (Status: 301) [Size: 227] [--> https://www.google.com/deleted]
design (Status: 301) [Size: 226] [--> https://www.google.com/design]
desktop (Status: 301) [Size: 227] [--> https://www.google.com/desktop]
developer (Status: 301) [Size: 229] [--> https://www.google.com/developer]
developers (Status: 301) [Size: 230] [--> https://www.google.com/developers]
devices (Status: 301) [Size: 227] [--> https://www.google.com/devices]
device (Status: 302) [Size: 242] [--> https://accounts.google.com/device]
docs (Status: 301) [Size: 224] [--> https://www.google.com/docs]
domains (Status: 301) [Size: 227] [--> https://www.google.com/domains]
doubleclick (Status: 301) [Size: 231] [--> https://www.google.com/doubleclick]
downloads (Status: 301) [Size: 229] [--> https://www.google.com/downloads]
ebooks (Status: 301) [Size: 226] [--> https://www.google.com/ebooks]
edu (Status: 301) [Size: 223] [--> https://www.google.com/edu]
education (Status: 301) [Size: 229] [--> https://www.google.com/education]
enterprise (Status: 301) [Size: 230] [--> https://www.google.com/enterprise]
errors (Status: 301) [Size: 226] [--> https://www.google.com/errors]
favicon.ico (Status: 301) [Size: 231] [--> https://www.google.com/favicon.ico]
files (Status: 301) [Size: 225] [--> https://www.google.com/files]
fi (Status: 302) [Size: 0] [--> https://fi.google.com/]
firefox (Status: 301) [Size: 226] [--> https://www.google.com/firefox]
finance (Status: 301) [Size: 0] [--> https://google.com/finance]
flights (Status: 301) [Size: 0] [--> https://google.com/flights]
fonts (Status: 301) [Size: 225] [--> https://www.google.com/Fonts]
fotos (Status: 301) [Size: 225] [--> https://www.google.com/fotos]
friends (Status: 301) [Size: 227] [--> https://www.google.com/friends]
forms (Status: 301) [Size: 225] [--> https://www.google.com/forms]
fusion (Status: 301) [Size: 226] [--> https://www.google.com/fusion]
games (Status: 301) [Size: 225] [--> https://www.google.com/games]
geo (Status: 301) [Size: 223] [--> https://www.google.com/geo]
get (Status: 301) [Size: 223] [--> https://www.google.com/get]
goto (Status: 301) [Size: 223] [--> http://www.google.com/goto]
grants (Status: 301) [Size: 226] [--> https://www.google.com/grants]
green (Status: 301) [Size: 225] [--> https://www.google.com/green]
groups (Status: 301) [Size: 226] [--> https://www.google.com/groups]
health (Status: 301) [Size: 226] [--> https://www.google.com/health]
hire (Status: 301) [Size: 224] [--> https://www.google.com/hire]
history (Status: 301) [Size: 0] [--> https://google.com/history]
home (Status: 301) [Size: 224] [--> https://www.google.com/home]
homepage (Status: 301) [Size: 228] [--> https://www.google.com/homepage]
hosted (Status: 301) [Size: 229] [--> https://www.google.com/hosted]
hotels (Status: 301) [Size: 226] [--> https://www.google.com/hotels]
ideas (Status: 301) [Size: 225] [--> https://www.google.com/ideas]
ie (Status: 301) [Size: 222] [--> https://www.google.com/ie]
ig (Status: 301) [Size: 222] [--> https://www.google.com/ig]
images (Status: 301) [Size: 225] [--> http://www.google.com/images]
inbox (Status: 301) [Size: 225] [--> https://www.google.com/inbox]
index.html (Status: 301) [Size: 229] [--> https://www.google.com/index.html]
jigsaw (Status: 301) [Size: 226] [--> https://www.google.com/jigsaw]
jobs (Status: 301) [Size: 224] [--> https://www.google.com/jobs]
js (Status: 301) [Size: 222] [--> https://www.google.com/js]
jump (Status: 301) [Size: 224] [--> https://www.google.com/jump]

```

Figure 9: Gobuster Directory Enumeration - Discovered Paths

Continued Gobuster results listing additional discovered directories/endpoints such as dashboard, developer, devices, docs, and favicon.ico, each with their corresponding HTTP status code and size, useful for mapping the attack surface of a web application.

```

wml (Status: 301) [Size: 222] [--> http://www.google.com/wml/]
work (Status: 301) [Size: 224] [--> https://www.google.com/work]
xhtml (Status: 301) [Size: 224] [--> http://www.google.com/xhtml]
Progress: 4613 / 4613 (100.00%)
=====
Finished
=====
--(akshat@kali)-[~]
_$ cat result.txt | grep -iE "config|backup|admin|login|.git|api|debug|password|secret|test|old|bak"
apis (Status: 301) [Size: 224] [--> https://www.google.com/apis]
password (Status: 301) [Size: 228] [--> https://www.google.com/password]
passwords (Status: 301) [Size: 229] [--> https://www.google.com/passwords]

--(akshat@kali)-[~]
_$ curl https://www.google.com/apis
<HTML><HEAD><meta http-equiv="content-type" content="text/html; charset=utf-8">
<TITLE>301 Moved</TITLE></HEAD><BODY>
<H1>301 Moved</H1>
The document has moved
<A HREF="https://code.google.com/">here</A>.
</BODY></HTML>

--(akshat@kali)-[~]
_$ curl https://www.google.com/password
<HTML><HEAD><meta http-equiv="content-type" content="text/html; charset=utf-8">
<TITLE>301 Moved</TITLE></HEAD><BODY>
<H1>301 Moved</H1>
The document has moved
<A HREF="http://support.google.com/accounts?prereset_pw_m">here</A>.
</BODY></HTML>

--(akshat@kali)-[~]
_$ nslookup google.com
Server: 10.16.177.22
Address: 10.16.177.22#53

Non-authoritative answer:
Name: google.com
Address: 192.178.177.113
Name: google.com
Address: 192.178.177.101
Name: google.com
Address: 192.178.177.102
Name: google.com
Address: 192.178.177.138
Name: google.com
Address: 192.178.177.100
Name: google.com
Address: 192.178.177.139
Name: google.com
Address: 2404:6800:4007:835::200e

```

Figure 10: Filtering Results and DNS/HTTP Verification

Completed Gobuster scan filtered with grep for sensitive keywords (config, admin, password, api, etc.), followed by curl requests to verify the discovered 'apis' and 'password' endpoints, and an nslookup query resolving google.com to its IP addresses.

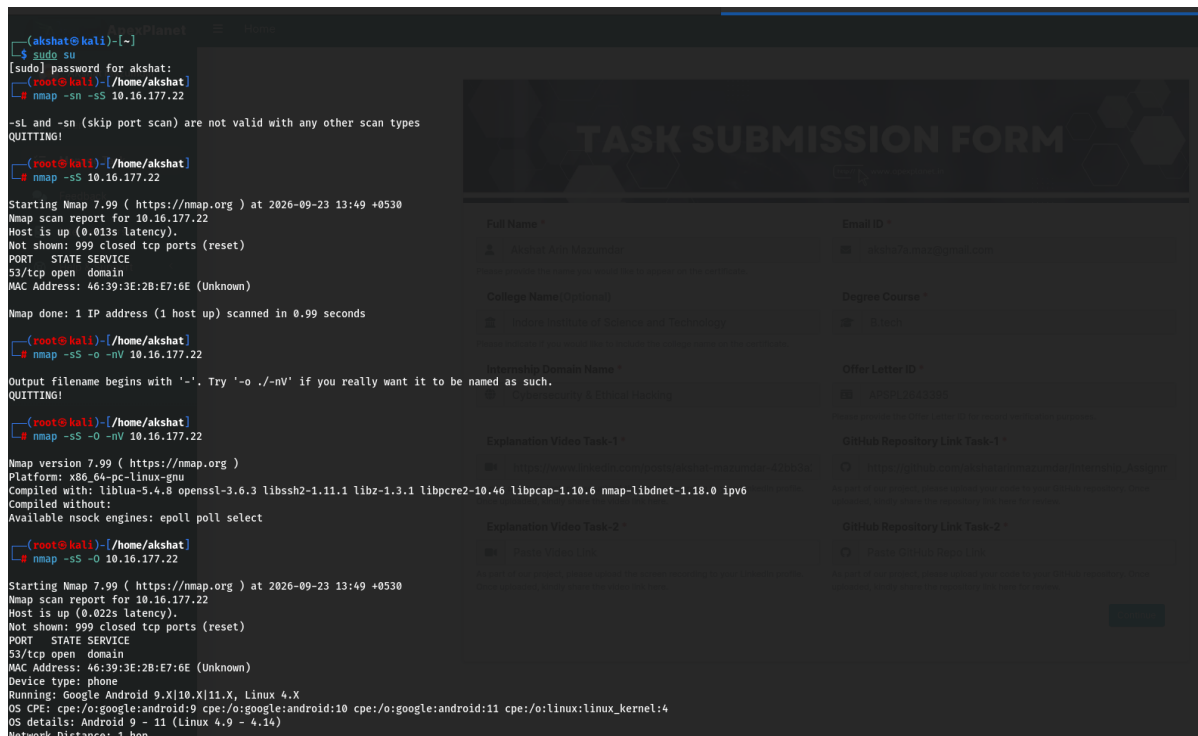


Figure 11: Nmap SYN Scan and OS Fingerprinting

Nmap SYN scan (-sS) against the target host revealing an open DNS port (53/tcp), followed by an OS detection scan (-O) that fingerprinted the device as an Android-based host, illustrating how Nmap can reveal both open services and underlying device details.

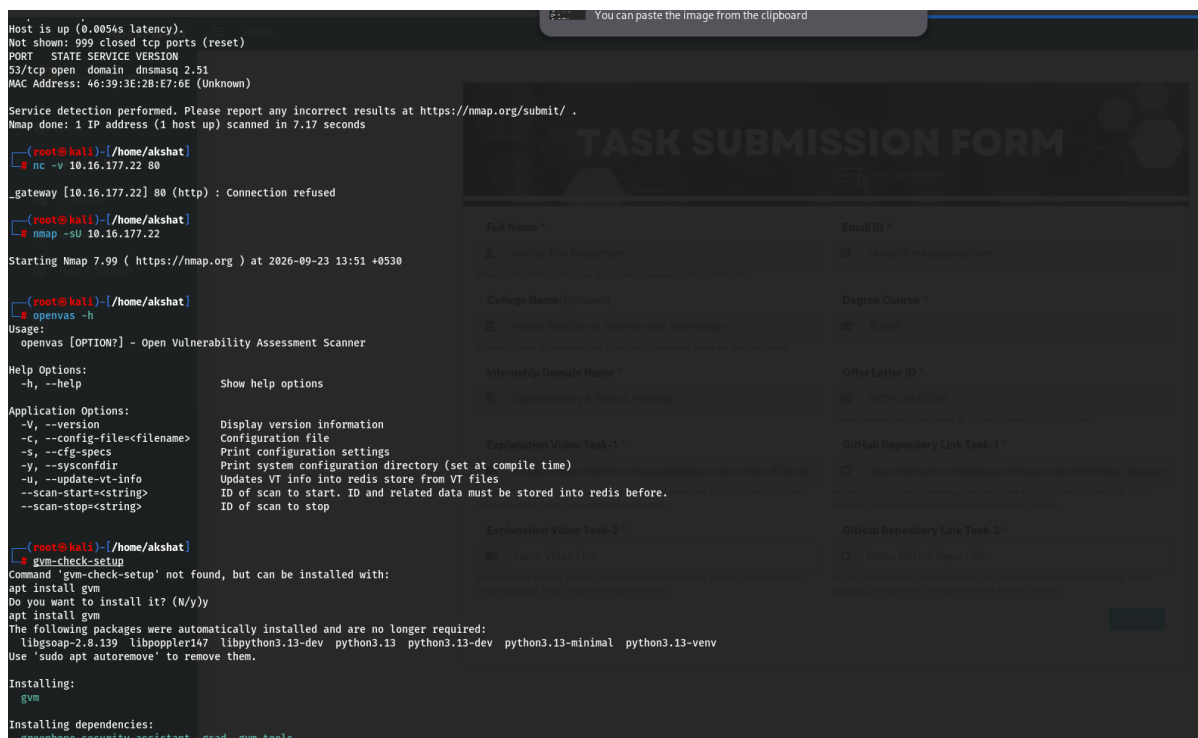


Figure 12: Service Detection, UDP Scan, and OpenVAS/GVM Preparation

Nmap service/version detection identifying dnsmasq 2.51 on port 53, a refused Netcat connection test on port 80, initiation of a UDP scan (-sU), the OpenVAS CLI help menu, and the beginning of the GVM package installation process required for vulnerability scanning.

```

You can paste the image from the clipboard
Use 'sudo apt autoremove' to remove them.

Installing:
gvm

Installing dependencies:
greenbone-security-assistant gsad gvm-tools

Summary:
Upgrading: 0, Installing: 4, Removing: 0, Not Upgrading: 473
Download size: 2,482 kB
Space needed: 15.9 MB / 36.0 GB available

Continue? [y/n] y
Get:1 http://kali.download/kali kali-rolling/non-free amd64 greenbone-security-assistant all 26.17.0-0kali1 [2,176 kB]
Get:2 http://mirrors.estio.network/kali kali-rolling/main amd64 gvm-tools all 25.4.6-1 [153 kB]
Get:3 http://kali.download/kali kali-rolling/main amd64 gsad amd64 24.16.0-1 [141 kB]
Get:4 http://kali.download/kali kali-rolling/main amd64 gvm all 25.04.3 [12.1 kB]
Fetched 2,482 kB in 7s (354 kB/s)
Selecting previously unselected package greenbone-security-assistant.
(Reading database... 609299 files and directories currently installed.)
Preparing to unpack ./greenbone-security-assistant_26.17.0-0kali1_all.deb...
Unpacking greenbone-security-assistant (26.17.0-0kali1)...
Selecting previously unselected package gsad.
Preparing to unpack ./gsad_24.16.0-1_amd64.deb...
Unpacking gsad (24.16.0-1)...
Selecting previously unselected package gvm.
Preparing to unpack ./archives/gvm_25.04.3_all.deb...
Unpacking gvm (25.04.3)...
Selecting previously unselected package gvm-tools.
Preparing to unpack ./gvm-tools_25.4.6-1_all.deb...
Unpacking gvm-tools (25.4.6-1)...
Setting up greenbone-security-assistant (26.17.0-0kali1)...
Setting up gsad (24.16.0-1)...
gsad.service is a disabled or a static unit, not starting it.
Setting up gvm (25.04.3)...
Setting up gvm-tools (25.4.6-1)...
Processing triggers for man-db (2.13.1-1)...
Processing triggers for kali-menu (2026.3.2)...
Scanning processes...
Scanning processor microcode...
Scanning linux images...

Running kernel seems to be up-to-date.

The processor microcode seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

```

Figure 13: GVM Package Installation Summary

APT package manager output showing the successful installation of the Greenbone Vulnerability Management stack - greenbone-security-assistant, gsad, gvm, and gvm-tools - which together provide the scanning engine and web interface for vulnerability assessment.

```

zsh: no such file or directory: https://10.16.177.22

(root@kali)-[/home/akshat]
└─# grep -i "password" /var/log/gvm/gvm-setup.log
grep: /var/log/gvm/gvm-setup.log: No such file or directory

(root@kali)-[/home/akshat]
└─# history | grep -i password
71  grep -i "password" /var/log/gvm/gvm-setup.log

(root@kali)-[/home/akshat]
└─# gvmc --user=admin --new-password=1234

(root@kali)-[/home/akshat]
└─# sudo find *gvm* -path *log* 2>/dev/null

(root@kali)-[/home/akshat]
└─# gvmc | grep password

(root@kali)-[/home/akshat]
└─# gvmc --get-users

(root@kali)-[/home/akshat]
└─# gvmc --user=admin --new-password=1234

(root@kali)-[/home/akshat]
└─# gvmc --create-user=admin --password=1234

(root@kali)-[/home/akshat]
└─# gvm-stop
[+] Stopping GVM services
o gsad.service - Greenbone Security Assistant daemon (gsad)
   Loaded: loaded (/usr/lib/systemd/system/gsad.service; disabled; preset: disabled)
   Active: inactive (dead)
   Docs: man:gsad(8)
        https://www.greenbone.net

Sep 23 14:33:53 kali gsad[29934]: gsad gmp:WARNING:2026-09-23 09h03.53 utc:29934: Authentication failure for 'admin' from 127.0.0.1. Status was 2.
Sep 23 14:33:55 kali gsad[29934]: gsad gmp:WARNING:2026-09-23 09h03.55 utc:29934: Authentication failure for 'admin' from 127.0.0.1. Status was 2.
Sep 23 14:34:21 kali gsad[29934]: gsad gmp:WARNING:2026-09-23 09h04.21 utc:29934: Authentication failure for 'admin' from 127.0.0.1. Status was 2.
Sep 23 14:34:22 kali gsad[29934]: gsad gmp:WARNING:2026-09-23 09h04.22 utc:29934: Authentication failure for 'admin' from 127.0.0.1. Status was 2.
Sep 23 14:41:42 kali gsad[29934]: gsad gmp:WARNING:2026-09-23 09h11.42 utc:29934: Authentication failure for 'admin' from 127.0.0.1. Status was 2.
Sep 23 14:41:43 kali gsad[29934]: gsad gmp:WARNING:2026-09-23 09h11.43 utc:29934: Authentication failure for 'admin' from 127.0.0.1. Status was 2.
Sep 23 14:41:58 kali gsad[29934]: gsad gmp:WARNING:2026-09-23 09h11.58 utc:29934: Authentication failure for 'admin' from 127.0.0.1. Status was 2.
Sep 23 14:44:58 kali systemd[1]: Stopping gsad.service - Greenbone Security Assistant daemon (gsad)...
Sep 23 14:44:58 kali systemd[1]: gsad.service: Deactivated successfully.
Sep 23 14:44:58 kali systemd[1]: Stopped gsad.service - Greenbone Security Assistant daemon (gsad).

o gvmc.service - Greenbone Vulnerability Manager daemon (gvmc)
   Loaded: loaded (/usr/lib/systemd/system/gvmc.service; disabled; preset: disabled)
   Active: inactive (dead)
   Docs: man:gvmc(8)

```

Figure 14: GVM Password Reset and Service Shutdown Logs

Resetting the GVM administrator password (gvmc --user=admin --new-password), followed by the gvm-stop command and system logs showing prior GSAD authentication failures, which were used to diagnose and correct the login credentials.

```

Sep 23 14:44:58 kali systemd[1]: Stopping notus-scanner.service - Notus Scanner...
Sep 23 14:44:58 kali systemd[1]: notus-scanner.service: Deactivated successfully.
Sep 23 14:44:58 kali systemd[1]: Stopped notus-scanner.service - Notus Scanner.

(root@kali)-[/home/akshat]
└─# gvm-start
[+] Please wait for the GVM services to start.
[+] You might need to refresh your browser once it opens.
[+] Web UI (Greenbone Security Assistant): https://127.0.0.1:9392

o gsad.service - Greenbone Security Assistant daemon (gsad)
   Loaded: loaded (/usr/lib/systemd/system/gsad.service; disabled; preset: disabled)
   Active: active (running) since Wed 2026-09-23 14:45:12 IST; 7ms ago
   Invocation: 6fdb4e2df3fa40f3bfe2807aa448d076
   Docs: man:gsad(8)
        https://www.greenbone.net

   Main PID: 58806 (gsad)
     Tasks: 1 (limit: 18072)
    Memory: 2.7M (peak: 2.7M)
       CPU: 12ms
    CGroup: /system.slice/gsad.service
            └─58806 /usr/sbin/gsad --foreground --listen 127.0.0.1 --port 9392
            └─58811 /usr/sbin/gsad --foreground --listen 127.0.0.1 --port 9392

Sep 23 14:45:12 kali systemd[1]: Starting gsad.service - Greenbone Security Assistant daemon (gsad)...
Sep 23 14:45:12 kali systemd[1]: Started gsad.service - Greenbone Security Assistant daemon (gsad).
Sep 23 14:45:12 kali gsad[58806]: gsad main: INFO:2026-09-23 09h15.12 utc:58806: Starting GSAD version 24.16.0-git
Sep 23 14:45:12 kali gsad[58811]: gsad main: INFO:2026-09-23 09h15.12 utc:58811: Starting HTTP server on 127.0.0.1 and port 9392
Sep 23 14:45:12 kali gsad[58806]: gsad main: INFO:2026-09-23 09h15.12 utc:58806: Starting HTTPS server on 127.0.0.1 and port 9392
Sep 23 14:45:12 kali gsad[58806]: gsad main:CRITICAL:2026-09-23 09h15.12 utc:58806: Binding to port 9392 failed.
Sep 23 14:45:12 kali gsad[58811]: gsad main: INFO:2026-09-23 09h15.12 utc:58811: gsad started successfully
Sep 23 14:45:12 kali systemd[1]: gsad.service: Main process exited, code=exited, status=1/FAILURE
Sep 23 14:45:12 kali systemd[1]: gsad.service: Failed with result 'exit-code'.

o gvmc.service - Greenbone Vulnerability Manager daemon (gvmc)
   Loaded: loaded (/usr/lib/systemd/system/gvmc.service; disabled; preset: disabled)
   Active: active (running) since Wed 2026-09-23 14:45:07 IST; 5s ago
   Invocation: 130139134daa42efb72374b967e24585
   Docs: man:gvmc(8)
   Process: 58757 ExecStart=/usr/sbin/gvmc --osp-vt-update=/run/ospd/ospd-openvas.sock --listen-group=_gvm (code=exited, status=0/SUCCESS)
   Main PID: 58759 (gvmc)
     Tasks: 3 (limit: 18072)
    Memory: 7.5M (peak: 10.7M)
       CPU: 78ms
    CGroup: /system.slice/gvmc.service
            └─58759 "gvmc: Waiting" --osp-vt-update=/run/ospd/ospd-openvas.sock --listen-group=_gvm
            └─58789 "gvmc: Synchron" --osp-vt-update=/run/ospd/ospd-openvas.sock --listen-group=_gvm
            └─58796 "gvmc: Syncing" --osp-vt-update=/run/ospd/ospd-openvas.sock --listen-group=_gvm

Sep 23 14:45:06 kali systemd[1]: Starting gvmc.service - Greenbone Vulnerability Manager daemon (gvmc)...
Sep 23 14:45:06 kali systemd[1]: gvmc.service: Can't open PID file '/run/gvmc/gvmc.pid' (yet?) after start: No such file or directory

```

Figure 15: GVM Services Restart and Confirmation

Confirmation that the Notus scanner service was stopped and the GVM stack was restarted using gvm-start, with the Greenbone Security Assistant (gsad) and Greenbone Vulnerability Manager (gvmd) services both reporting an active (running) status, making the web-based vulnerability scanner ready for use.

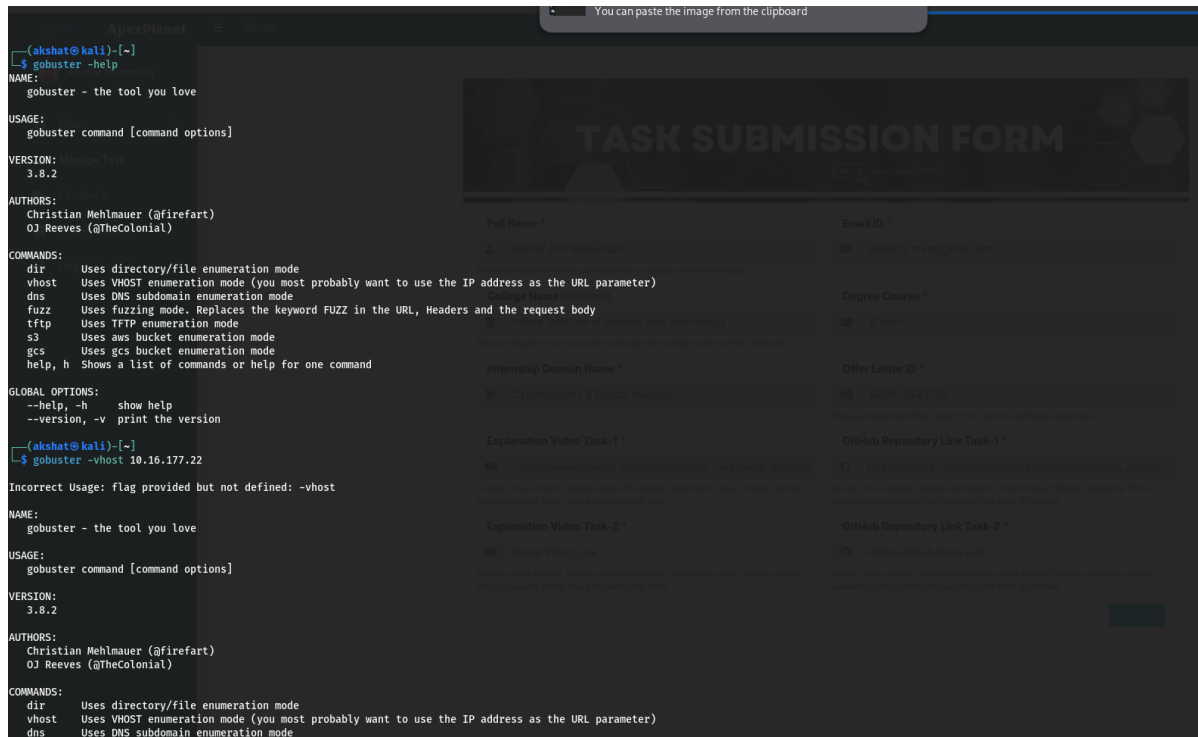


Figure 16: Gobuster Help Menu and VHOST Mode Error

The Gobuster help menu listing its available enumeration modes (dir, vhost, dns, fuzz, tftp, s3, gcs), followed by an example of an incorrect command-line flag (-vhost) being rejected, demonstrating proper command syntax verification before running a scan.

```

GLOBAL OPTIONS:
  --help, -h      show help
  --version, -v   print the version
2026/09/23 13:23:43 flag provided but not defined: -vhost

(akshat@kali)-[~]
$ gobuster -fuzz google.com

Incorrect Usage: flag provided but not defined: -fuzz

NAME:
  gobuster - the tool you love

USAGE:
  gobuster command [command options]

VERSION:
  3.8.2

AUTHORS:
  Christian Mehlmauer (@firefart)
  OJ Reeves (@TheColonial)

COMMANDS:
  dir      Uses directory/file enumeration mode
  vhost    Uses VHOST enumeration mode (you most probably want to use the IP address as the URL parameter)
  dns      Uses DNS subdomain enumeration mode
  fuzz     Uses fuzzing mode. Replaces the keyword FUZZ in the URL, Headers and the request body
  tftp     Uses TFTP enumeration mode
  s3       Uses aws bucket enumeration mode
  gcs      Uses gcs bucket enumeration mode
  help, h  Shows a list of commands or help for one command

GLOBAL OPTIONS:
  --help, -h      show help
  --version, -v   print the version
2026/09/23 13:24:09 flag provided but not defined: -fuzz

(akshat@kali)-[~]
$ gobuster google.com

No help topic for 'google.com'

(akshat@kali)-[~]
$ gobuster dir -u google.com -w /usr/share/wordlist/dirb/wordlists/dirb/common.txt

2026/09/23 13:27:33 wordlist file "/usr/share/wordlist/dirb/wordlists/dirb/common.txt" does not exist: stat /usr/share/wordlist/dirb/wordlists/dirb/common.txt: no such file or directory

(akshat@kali)-[~]

```

Figure 17: Gobuster Fuzzing Mode and Wordlist Path Error

An attempt to use an invalid `-fuzz` flag, followed by a directory scan attempt that failed because the specified wordlist path was incorrect, highlighting the importance of validating file paths (e.g., `dirb/common.txt`) before executing enumeration tools.

```

(akshat@kali)-[~]
$ gobuster dir -u google.com -w /usr/share/wordlists/dirb/common.txt

=====
Gobuster v3.8.2
by OJ Reeves (@TheColonial) & Christian Mehlmauer (@firefart)
=====
[*] Url: http://google.com
[*] Method: GET
[*] Threads: 10
[*] Wordlist: /usr/share/wordlists/dirb/common.txt
[*] Negative Status codes: 404
[*] User Agent: gobuster/3.8.2
[*] Timeout: 10s
=====
Starting gobuster in directory enumeration mode
=====
2002 (Status: 301) [Size: 224] [--> https://www.google.com/2002]
2003 (Status: 301) [Size: 224] [--> https://www.google.com/2003]
2004 (Status: 301) [Size: 224] [--> https://www.google.com/2004]
2006 (Status: 301) [Size: 224] [--> https://www.google.com/2006]
2009 (Status: 301) [Size: 224] [--> https://www.google.com/2009]
2010 (Status: 301) [Size: 224] [--> https://www.google.com/2010]
2008 (Status: 301) [Size: 224] [--> https://www.google.com/2008]
2007 (Status: 301) [Size: 224] [--> https://www.google.com/2007]
2009 (Status: 301) [Size: 224] [--> https://www.google.com/2009]
2013 (Status: 301) [Size: 224] [--> https://www.google.com/2013]
2014 (Status: 301) [Size: 224] [--> https://www.google.com/2014]
2012 (Status: 301) [Size: 224] [--> https://www.google.com/2012]
2011 (Status: 301) [Size: 224] [--> https://www.google.com/2011]
A (Status: 301) [Size: 221] [--> https://www.google.com/A]
a (Status: 301) [Size: 221] [--> https://www.google.com/a]
about (Status: 301) [Size: 225] [--> https://www.google.com/about]
accessibility (Status: 301) [Size: 233] [--> https://www.google.com/accessibility]
account (Status: 301) [Size: 227] [--> https://www.google.com/account]
action (Status: 301) [Size: 225] [--> http://www.google.com/action]
activity (Status: 301) [Size: 0] [--> https://google.com/activity]
accounts (Status: 302) [Size: 237] [--> https://accounts.google.com/]
ads (Status: 301) [Size: 223] [--> https://www.google.com/ads]
advertisers (Status: 301) [Size: 225] [--> https://www.google.com/advertisers]
advertising (Status: 301) [Size: 231] [--> https://www.google.com/advertising]
advertise (Status: 301) [Size: 229] [--> https://www.google.com/advertise]
adview (Status: 301) [Size: 225] [--> http://www.google.com/adview]
advanced_search (Status: 301) [Size: 234] [--> http://www.google.com/advanced_search]
africa (Status: 301) [Size: 226] [--> https://www.google.com/africa]
analytics (Status: 301) [Size: 229] [--> https://www.google.com/analytics]
android (Status: 301) [Size: 227] [--> https://www.google.com/android]
alerts (Status: 301) [Size: 226] [--> https://www.google.com/alerts]
answers (Status: 301) [Size: 227] [--> https://www.google.com/answers]
apis (Status: 301) [Size: 224] [--> https://www.google.com/apis]
appliance (Status: 301) [Size: 229] [--> https://www.google.com/appliance]

```

Figure 18: Gobuster Directory Scan - Successful Execution

A corrected Gobuster directory scan command using the valid wordlist path (/usr/share/wordlists/dirb/common.txt), successfully enumerating multiple valid endpoints on google.com along with their HTTP status codes and response sizes.

```

city (Status: 301) [Size: 224] [--> https://www.google.com/city]
collection (Status: 301) [Size: 0] [--> https://google.com/collection]
commerce (Status: 301) [Size: 228] [--> https://www.google.com/commerce]
compare (Status: 301) [Size: 227] [--> https://www.google.com/compare]
company (Status: 301) [Size: 227] [--> https://www.google.com/company]
cm (Status: 301) [Size: 0] [--> http://campaignmanager.google.com/cm]
contact (Status: 301) [Size: 227] [--> https://www.google.com/contact]
contacts (Status: 301) [Size: 228] [--> https://www.google.com/contacts]
contributor (Status: 301) [Size: 231] [--> https://www.google.com/contributor]
creditcards (Status: 301) [Size: 231] [--> https://www.google.com/creditcards]
crossdomain.xml (Status: 301) [Size: 235] [--> https://www.google.com/crossdomain.xml]
cse (Status: 301) [Size: 223] [--> https://www.google.com/cse]
custom (Status: 301) [Size: 226] [--> https://www.google.com/custom]
customers (Status: 301) [Size: 229] [--> https://www.google.com/customers]
cr (Status: 400) [Size: 16]
dashboard (Status: 301) [Size: 229] [--> https://www.google.com/dashboard]
deleted (Status: 301) [Size: 227] [--> https://www.google.com/deleted]
design (Status: 301) [Size: 226] [--> https://www.google.com/design]
desktop (Status: 301) [Size: 227] [--> https://www.google.com/desktop]
developer (Status: 301) [Size: 229] [--> https://www.google.com/developer]
developers (Status: 301) [Size: 230] [--> https://www.google.com/developers]
devices (Status: 301) [Size: 227] [--> https://www.google.com/devices]
device (Status: 302) [Size: 243] [--> https://accounts.google.com/device]
docs (Status: 301) [Size: 224] [--> https://www.google.com/docs]
domains (Status: 301) [Size: 227] [--> https://www.google.com/domains]
doubleclick (Status: 301) [Size: 231] [--> https://www.google.com/doubleclick]
downloads (Status: 301) [Size: 229] [--> https://www.google.com/downloads]
ebooks (Status: 301) [Size: 226] [--> https://www.google.com/ebooks]
edu (Status: 301) [Size: 223] [--> https://www.google.com/edu]
education (Status: 301) [Size: 229] [--> https://www.google.com/education]
enterprise (Status: 301) [Size: 230] [--> https://www.google.com/enterprise]
errors (Status: 301) [Size: 226] [--> https://www.google.com/errors]
favicon.ico (Status: 301) [Size: 231] [--> https://www.google.com/favicon.ico]
files (Status: 301) [Size: 225] [--> https://www.google.com/files]
finance (Status: 301) [Size: 0] [--> https://google.com/finance]
firefox (Status: 301) [Size: 226] [--> http://www.google.com/firefox]
fl (Status: 302) [Size: 0] [--> https://fl.google.com/]
flights (Status: 301) [Size: 0] [--> https://google.com/flights]
fonts (Status: 301) [Size: 225] [--> https://www.google.com/fonts]
forms (Status: 301) [Size: 225] [--> https://www.google.com/forms]
fotos (Status: 301) [Size: 225] [--> https://www.google.com/fotos]
friends (Status: 301) [Size: 227] [--> https://www.google.com/friends]
fusion (Status: 301) [Size: 226] [--> https://www.google.com/fusion]
games (Status: 301) [Size: 225] [--> https://www.google.com/games]
geo (Status: 301) [Size: 223] [--> https://www.google.com/geo]
get (Status: 301) [Size: 223] [--> https://www.google.com/get]
goto (Status: 301) [Size: 223] [--> http://www.google.com/goto]
grants (Status: 301) [Size: 226] [--> https://www.google.com/grants]
green (Status: 301) [Size: 225] [--> https://www.google.com/green]
groups (Status: 301) [Size: 226] [--> https://www.google.com/groups]
health (Status: 301) [Size: 226] [--> https://www.google.com/health]
hire (Status: 301) [Size: 224] [--> https://www.google.com/hire]
history (Status: 301) [Size: 0] [--> https://google.com/history]

```

Figure 19: Gobuster Directory Scan - Extended Results

Further enumerated endpoints from the Gobuster scan (e.g., dashboard, deleted, design, devices, docs, ebooks, favicon.ico), providing a broader view of the target's directory structure and helping identify potentially interesting or sensitive paths for further investigation.

5. Conclusion / Learnings

This task provided practical, end-to-end exposure to the core stages of a network security assessment - reconnaissance, scanning, vulnerability assessment, traffic analysis, and controlled stress testing. Working with Nmap deepened the understanding of how open ports, running services, and operating systems can be fingerprinted remotely. Configuring and troubleshooting the OpenVAS/GVM stack (resolving feed locks, service failures, and authentication issues) highlighted the operational complexity behind automated vulnerability scanners and reinforced the importance of methodical debugging. Capturing traffic with Wireshark showed how packet-level visibility supports the verification of scanning and flood activity, while the hping3 SYN flood exercise illustrated, in a safe and controlled manner, how excessive SYN traffic can be generated and observed - reinforcing the rationale behind SYN-flood mitigations and firewall (iptables) filtering. Finally, the Gobuster directory enumeration exercise demonstrated how unlisted or forgotten endpoints can expose additional attack surface on a web application, and underscored the value of correct tool syntax and wordlist selection for effective enumeration.

Overall, the exercise strengthened both the technical proficiency and the analytical mindset required for ethical hacking and network security assessments, and emphasised that these tools must always be used strictly within authorised, controlled environments.